CHAPTER – 10

STEEL WORK

10.1 MATERIALS

10.1.1 STEEL

Finished steel shall be well and cleanly rolled to the dimensions and weight subject to permissible tolerances as per IS:1852. Finished material shall be free from cracks, surface flaws, laminations, imperfect and rough edges and any other defects. They shall be free from rust, scaling and putting and shall be well protected. For structural purpose following varieties of steel only should be used unless otherwise specified.

(a) ST 42-S conforming to IS:226: For all types of structures (riveted or bolted) including those subjected to dynamic loading and where fatigue, wide fluctuation of stresses, reversal of stresses are involved like in crane gantry girders, road and rail bridges etc. It is suitable for welding only if the thickness of the materials does not exceed 20 mm.

(b) S.T. 42-W conforming to IS 2062: This is fusion welding quality steel used for structures like crane gantry girders, road and rail bridges etc.

(c) S.T. 42-O conforming to IS 1977: This is ordinary quality steel which shall be used only for structures not subjected to dynamic loading, where welding is not employed, structures not situated in earth quake zones and structures the design of which has not been based on plastic theory.

(d) S.T.. 32-O conforming to IS 1977: This is ordinary quality steel which shall be used only for doors, window frames, window bars, grills, steel gates, hand railing, builders hardware, fencing posts, tie bars etc.

(e) All structural steel work shall conform to code of practice for use of structural steel in general building construction IS:800.

(f) All structural steel and electrodes shall comply in all respects with Indian Standard Specification for structural steel.

10.2 PAINTING

Surfaces to be painted, oiled or otherwise treated shall be dry and thoroughly cleaned to remove all loose scales and loose rust.

Inaccessible surfaces after shop assembly shall receive full specified protective treatment before assembly except sealed hollow sections. The part of steel member to be encased in concrete shall not be painted or oiled. Prior to placing any steel member in position or taken out of workshop a priming coat of approved steel primer i.e., Red oxide zinc chromate primer conforming to IS:2074 should be applied.

10.3 WORKMANSHIP

All workmanship shall be of acceptable standard in every respect, and accuracy should be ensured so that all parts will fit together properly on erection.

All ends shall be cut true to planes. They must fit the abutting surfaces closely.

All stiffeners shall bear tightly at both ends.

All butt ends of compression members shall be in close contact throughout the area of the joint.

All holes in plates and sections between 12.7 mm and 19 mm thick shall be punched to such a diameter that 3.18 mm of metal is left all around the hole to be cleaned out to correct size by reamer.

The base connection shall be provided as shown in drawings and accurate workmanship shall be ensured to provide the connections.

Figured dimensions on the drawings shall be taken.

10.4 ERECTION

(a) All the plants and equipments used for erection should be of adequate capacity and contractor has to furnish details of same to Engineer In-Charge well before the programmed date for erection and get his clearance/approval.

(b) During erection, the work shall be securely braced and fastened temporarily to provide safety against all erection stresses etc. No permanent welding shall be done until proper alignment has been obtained.

(c) Any parts which do not fit accurately or which are not in accordance with the drawings and specifications shall be liable to rejection and if rejected shall at once be made good.

(d) Engineer-in-Charge shall have full liberty at all reasonable times to enter the contractor's premises for the purpose of inspecting the work and no work shall be taken down, painted or dispatched until it has been

inspected and passed. The contractor shall supply free of charge all labour and tools required for testing of work.

NOTE : On any straight weld the first run shall not ordinarily be deposited with a larger diameter electrode than 4 mm. For subsequent run the electrodes shall not be increased by more than two electrodes between consecutive runs.

10.5 WELDING OPERATIONS

The contractor shall ensure that each welding operator employed on fabrication or erection is an efficient and dependable welder, who has passed qualifying tests on the types of welds which he will be called upon to make. Sample tests shall have to be given by the contractor to the entire satisfaction of the Engineer-in-Charge.

10.6 WELDING PROCESS

A) The work shall be positioned for downward welding whereever possible.

B) Arc length, voltage and amperage shall be suited to the thickness of material, type of groove and other circumstances of the work. The welding current and electrode sizes for different types of joints shall be as per IS:9595.

C) The sequence of welding shall be such as to avoid undue distortion and minimize residual shrinkage stresses. Recommendation of IS:9595 shall be followed.

D) The electrode manipulation during welding shall be such as to ensure that:

1) The parent metal is in a fused stage when the filler metal makes contact with it.

2) The weld metal does not overflow upon any unfused parent metal forming overlappings.

3) The parent metal is not under-cut along the weld toes.

4) The flowing metal floats, the slag, the oxides, and the gas bubbles to the surface behind the advancing pool. In case any of these requirements is unattainable by manipulation, the current shall be adjusted or the electrode size changed.

Each time the arc is started the electrode shall be moved in such a way that the fusion of base metal at the starting point is assured. At the completion of a run the movement of electrode shall be slowed down to fill the arc crater.

After every interruption of the arc except at completion of a run, the arc shall be re-started ahead of the previous deposit and then moved back to fill the center or such alternative technique shall be used as will ensure complete filling of the crater, or complete fusion between the new and old deposit and the base metal at the point of junction, and result in continuity of weld. Before welding operation is completed, all traces of slag shall be removed from the deposit, by chipping if necessary, and the deposit and the adjoining base metal shall be wire brushed and cleaned at all points. The requirements shall apply not only to successive layers, but also to successive beads, and to the over lapping area wherever a junction is made on starting a new electrode.

5) The welds shall be free from cracks, discontinuity in welding and other defects such as (I) under-size, (ii) over-size, (iii) under-cutting and (iv) over-cutting in the case of fillet welds and defects (ii), (iii) & (iv) in the case of butt welds.

   All defective welds which shall be considered harmful to the structural strength shall be subjected to radiographic examination as described in IS:1182.

   In case of welded butt joints in steel of thickness upto 50 mm the weld joint shall be subjected to radiographic examination as described in IS:1182.

   All welds shall be cleaned of slag and other deposits after completion. Till the work is inspected and approved painting shall not be done. The surface to be painted shall be cleaned of spatter, rust, loose scale, oil and dirt.

   Inspection and testing of welding shall be as per IS:822.

10.7 Welding shall generally be done by electric process or as **per IS:816,** 7069 and 823-1964. Gas Welding shall only be resorted to using oxyacetylene flame with specific approval of the Engineer-in-Charge. Gas welding shall not be permitted to structural steel work.

10.8 WORKMANSHIP

10.8.1 PREPARATION OF FUSION FACES

(a) Fusion faces shall be cut by shearing machine or gas cutting and later dressed by filing or grinding, so that they shall be free from irregularities such as would interfere with the deposition of the specified size of weld to be the cause of defects. Fusion faces and the surrounding surfaces shall be free from heavy slag and free from oil, paints or any other substance which might affect the quality of the weld or impede the progress of welding.

(b) The parts to be welded shall be brought into as close contact as possible and the gap due to faulty workmanship or incorrect fit up shall not exceed 1.6 mm. If separation of 1.6 mm or more occurs locally the size of the fillet welds shall be increased at such positions by an amount equal to the width of the gap.

10.9 WELDING

SIZE OF ELECTRODES AND RUNS

The maximum dia of the electrodes for welding any work shall be as per IS:814 and appendix 'B' of IS:823 and in the following table whichever is least.

Average thickness of plate or section	**Maximum cut diameter of electrodes to be used**
Less than 5 mm	3.8 mm
5 mm upto but not including 8 mm	4 mm
8 mm upto but not including 10mm	5 mm
10 mm upto but not including 18 mm	6 mm
16 mm upto but not including 25 mm	9 mm
25 mm and over	9 mm

The parts to be welded shall be maintained to their correct position during welding. They shall be securely held in position by means of tackwelds, service bolts, clamps or rings before commencing welding so as to prevent any relative movement due to distortion, wind or any other cause. Step back mentioned should be used to avoid distortion.

The minimum leg length of a filler weld as deposited should be not less than the specified size and the throat thickness as deposited should be not less than that tabulated below:

THROAT THICKNESS OF FILLET

Angle between fusion faces	60° – 90°	91° -100°	101° -106°	107° -113°
Throat thickness in cm	0.70	0.65	0.60	0.55

In no case should a concave weld be deposited without the specific approval of the Engineer-in-Charge unless the leg length is increased above the specified, so that the resultant throat thickness is as great as would have been obtained by the deposition of fillet.

All welds shall be deposited in a pre-arranged order and sequence taking due account of the effects of distortion and shrinkage stresses.

After making each run of welding, all slag shall be removed and final run shall be protected by clean boiled linseed oil till approved.

The weld metal, as deposited shall be free from crack, slag, excessive porosity, cavities and other faults.

The weld metal shall be properly fused with the parent metal without over-lapping or series under-cutting at the tees of the weld.

The surfaces of the weld shall have a uniform and consistent contour and regular appearance.

In welds containing cracks, porosity or cavities in which the old metal tends to overlap in the parent metal without proper fusion the defective portions of the welds shall be cut out and re-welded. Where serious undercutting occurs additional weld metal shall be deposited to make good the reduction.

Peeling of welds shall be carried out only with the approval of the Engineer-in-Charge.

Immediately after de-slagging, inspection and approval, all site welds and the surrounding surfaces shall be painted to protect the metal with clean boiled linseed oil.

10.10 BUTT WELDS

The ends of butt joints shall be welded so as to provide the full throat thickness. The weld face shall be at all places proud of the surface of the parent metal but shall not exceed 3.18 mm.

10.11 SHOP DRAWINGS

The shop drawings indicating all the requisite details of structural steel based on contract drawings shall be submitted to the Engineer-in-Charge along with the necessary information for fabrication, erection, painting of structures etc, immediately after acceptance of the tender.

10.12 FABRICATION OF BUILT UP SECTION BY WELDING

The fabrication shall generally be done as specified in IS:800. Adequate care shall be taken while fabricating to ensure accuracy so that these can be assembled without being unduly packed, strained or unduly forced into position. The built up section shall be true and free from twist, kinks, buckles, or open

joints. Suitable templates of wood or metal shall be made to facilitate accurate fabrication. Unless approved in fabrication drawing no two pieces shall be welded or joined to make up for the required length of the member.

10.13 ERECTION AND FIXING IN POSITION PREFABRICATED TRUSSES AND COLUMNS

The prefabricated steel laced columns are to be strictly according to the details furnished in the drawing. All scaffolding, temporary supports, tools, trusses required for erecting, hoisting and fixing in position the columns and trusses shall be provided by the contractor at his own cost. The trusses and columns shall be erected and fixed in the position true to plumb and levels and lines in a workman-like manner as directed by the Engineer-in-Charge. The various components being erected shall be supported and held in position suitably till completion of erection. Any damage to the materials caused by the contractor shall be made good and rectified at his own cost to the entire satisfaction of the Engineer In-Charge.

10.14 STEEL DOORS, WINDOWS AND VENTILATORS

The doors, windows and ventilators shall be manufactured from uniform IS standard sections and or M.S. angles to suit sizes as given in the schedule of quantities. The weight of different rolled steel section used for the manufacture of doors, windows and ventilators, shall not be less than those specified in IS:103 or 1038 unless otherwise specified. The steel shall be of fusion welding quality St. 42 designation. The steel section shall conform in all respects to IS:4062. In case where the standard door and windows are of insufficient strength for the particular door, window, ventilator etc. suitable other M.S rolled or built up sections of sufficient strength shall be used. Necessary Mild Steel stiffeners are to be provided in doors wherever required for structural stability. The contractor shall produce detailed shop drawings for the doors, windows and ventilators showing the sizes and arrangement of the sections and units and system of operation for the approval of the Engineer-in-Charge and fabrication shall be commenced only after approval to these shop drawings.

Corners of doors, and ventilators shall be accurately mitred, jointed and fitted to produce flush joints and folded along the concealed line of contact. The corners of fixed and opening frame shall be electrically flash butt-welded to form a solid and true right angle and all frames shall be square and flat.

The weld shall thoroughly penetrate the metal resulting in complete fusion between frames. Mechanically jointed corners or those developed by the rods and/or superficial surface welding shall not be permitted.

All cut outs, recesses, mortising or milling operations required for hardware shall be accurately made and reinforced with backing plates as required to

ensure adequate strength of the connections. No field fabrication of frames shall be permitted.

Intermediate glazing bars shall be tenoned and riveted into the frames and inter-sections, the horizontal glazing bars shall pass through the vertical bars and the joints closed by hydraulic pressure.

The steel doors and windows shall be according to the specifications and design. The sizes of doors and windows shall be calculated so as to allow 1.24 cm clearance on all the four sides of opening to allow for easy fitting of doors, windows and ventilators. The actual sizes of doors, windows and ventilators shall not vary by more than 1.5 mm from those given in the design.

The thickness and type of glazing shall be as specified in the schedule. The glass panes shall be free from flaws, specks or bubbles and shall have square corners and straight loose in the frames.

The beading shall be of standard aluminium channel as shown on drawing and approved. The manufacturers shall be intimated in advance to drill holes for head screws. Beads shall be fixed with screws spaced not more than 10 cm from each corner.

The screws, nuts, washers, bolts, rivets, weather bars and any other miscellaneous fastening services shall be of steel. Samples of doors/windows/ ventilators with all fittings, locking arrangement peg stays, handles etc. shall be submitted for approval to the Engineer-in-Charge and got approved before taking up the manufacture of full quantity. No extra payment will be made towards the supply of same for approval.

The component parts of doors, windows, ventilators shall all be as per approved working drawings.

Composite units consist of a combination of two or more units of doors, windows and ventilators etc. as the case may be. The different units shall be coupled by using coupling sections. The coupling sections shall be made from M.S sheet 3.15 mm in thickness and 56 mm wide as per IS:1038 para 5.2 and these shall be fixed with bolts and nuts.

Wherever the doors, windows and ventilators shall be coupled with a coupling section mastic cement shall be applied between the junction to make the joint water tight.

All steel surfaces shall be thoroughly cleaned of rust, scale and dirt by pickling and marking. A shop priming coat of superior quality redoxide or equivalent shall then be given before despatch.

Unless otherwise specified the contractor's rate for doors, windows, ventilators shall be inclusive of supply of necessary fixtures, fittings, fastenings like lugs, beading, peg stays, handles, locking arrangement, screws, spring catch, applying a priming coat, glazing etc. complete.

10.15 REQUIREMENT OF WELDED JOINTS OF STEEL DOORS, WINDOWS, VENTILATORS AND COMPOSITE UNITS

(I) VISUAL INSPECTION TEST

When two opposite corners of the frame are cut, paint removed and inspected the joint shall conform to:

(a) Welds should have been made all along the place of meeting of the members and there is no tack welding.

(b) Welds are properly grounded.

(c) The joint is completely solid and no visible cavities.

(II) MICRO AND MACRO EXAMINATION

From the two opposite corners obtained for visual test, the flanges of the sections shall be cut with the help of a saw. The cut surface of the remaining portions shall be polished, etched and examined. The polished and etched faces of the weld and base metal shall be free from cracks and reasonably free from under cutting, over laps, gross porosity and entrapped slag.

(III) FILLET WELD TEST

The fillet weld in the remaining portion of the joint shall be fractured by hammering. The fractured surface shall be free from slag, inclusion porosity, crack penetration defects and fusion defects.

10.16 FIXING OF STEEL DOORS, WINDOWS & VENTILATORS

Steel doors and windows shall be stacked in upright position on level ground, preferably on wooden battens to keep them in true shape without damage.

Doors, windows and ventilators shall be fixed to masonry (inclusive of brick, RCC stone and marble) timber and steel work openings as shown in drawings in the following manner.

Openings may be flush or rebated as shown in the drawing. These openings may have rendered finish or a 'flair faced' finish (i.e., without rendering as in case of marble or stone facing). Where openings are flush and with a rendered finish a clearance of 1.25 cm shall be provided between the steel frame

and opening. In case of external masonry finish "fair faced" and with rebated jambs, a minimum 1.25 cm clearance between frame and opening shall be provided.

Steel work opening shall be so designed that the outer flange of the door, windows or ventilator frame section overlaps the steel surface by 10 mm.

Outer frame shall be provided with fixing holes centrally in the web of the sections in the position given in IS:1038. The materials for fixing frames to opening in different class of opening shall generally conform to IS:1038-1957.

10.17 FIXING PROCEDURE

MASONRY OPENINGS

(a) Fixing with lugs:

(i) Doors, windows and ventilators unit, shall not be 'built' in as the work proceeds but opening shall be left alround. The size of the opening shall first be checked and cleared of obstruction, if any.

The position of the unit of fixing holes shall be marked on the jamb. Necessary holes shall be made in the masonry and lugs not less than 10cm long 15 x 3 mm size fixed in cement concrete blocks 15 x 10 x 10 cm size of 1:3:6 mix (1 cement :3 coarse sand :6 graded stone aggregate 20 mm nominal size).

The frames of units shall be set in the openings by using wooden wedges at the jamb, head and sill, (wedges shall preferably be placed near the points where a glazing bar meets the frames) and be plumbed in position.

(ii) Then it lugs shall be fixed with the frame with 20 mm long and 7.3 mm dia G.I. Counter sunk machine screws and nuts. In case of flush opening which are rendered smooth wedges shall be removed and gap between unit and the jambs shall be filled with approved filling material.

(iii) In case of flush jamb with external 'fair faced' finish the gap between the opening and frame shall be filled with mastic from inside till it oozes out on external face. The oozing mastic shall be cleaned and flush pointed. The internal gap shall be filled with mastic to about 1/ 3rd depth and the rest with cement mortar.

(iv) In case of rebated jambs and jambs finished 'fair faced ' externally the mastic shall be freely applied to the inside channel of frame, jamb and sill, so as to ensure a water tight joint. After the unit is firmly fixed in position surplus mastic shall be cleaned and flush pointed.

(b) Fixing with screws and plugs

In RCC work where lugs cannot be embedded due to reinforcement bars etc. wooden plugs shall be fixed in RCC at proper position while laying concrete. Alternatively rawl plugs may be fixed in proper position and frames fixed to them with 60 mm galvanized wood screws. Alternatively approved anchore fastener may be used.

Wood Work Openings:

Wooden openings are normally rebated and mastic shall be applied to jambs, sill of a channel before fixing in position. The frame shall be set in opening using wooden wedges and fixed to the opening with 60 mm galvanized wood screws.

Extra timber fillets of hard wood to match the adjoining work shall also be provided around the frame to close the extra gap between opening and frame.

STEEL WORK OPENINGS

Before placing the unit frame in position mastic shall be applied and a Mild Steel or hard wood fillet shall be provided around the frame to close the extra gap between opening and frame. The unit shall then be fixed to the opening with fixing clips or with nuts and bolts as shown in the drawings or as directed by the Engineer In-Charge.

10.18 FIXING OF COMPOSITE UNITS

The fixing procedure for composite units shall generally be as described above except that:

(a) Where large units shall be formed by coupling individual units together (with coupling sections), the mullions and transoms shall be bedded in mastic to ensure water tightness. Mastic shall be applied liberally to the channels of the outside frame sections before assembly and after coupling. All oozing out mastic shall be cut out neatly.

(b) Mullions normally project 2.5 cm at head and sill and are fixed in pockets made into the masonry, timber or steel work opening. But where it is at cross joint with a transom the shorter coupling unit shall run through unbroken and other coupling unit shall be cut square to form a butt with other members.

10.19 COLLAPSIBLE GATES

These shall be from the approved manufacturers and shall be fabricated from the mild steel sections. The details of vertical channels, flat iron diagonals, top and bottom rails, balls bearings etc. shall be as per approved manufacturers specification. The bolts, nuts, locking arrangements, stoppers, handles, special

fittings like springs, catch and locks shall all be as specified in the description of item. The gate shall open and close smoothly and easily.

The height of the gate shall be measured as length of the vertical double channels and breadth from outside to outside of the end fixed double channels in open position of the gate. The area shall be measured in sq meters correct to two decimals.

10.20 M.S SHEET SLIDING SHUTTERS

These shall be manufactured as per approved shop drawings.

The shutters, MS angle iron frame work, diagonal braces, gusset plates, MS sheet coverings, top and bottom guide rails, pulleys, ball bearings., stoppers, hold fasts, other fittings and all shall be indicated in detail in shop drawings and got approved by Engineer-in-Charge prior to fabrication. The guide rails shall be sufficiently long and continued along the wall on both ends so that sliding shutter can rest in the wall portion giving full opening if required.

The details of guide rail fixing to floor by anchor bolts embedded in cement concrete, supporting of steel sections at top, damping arrangement to avoid shutters rolling back with the opening shall all be indicated in the shop drawing and got approved.

Two channel sections shall be suitably fixed vertically below the extreme clamps in the wall and floor to avoid the shutter from going out of the support at top and bottom.

Damages if any to the adjoining work while erecting/fixing the sliding door shall be made good to match the adjoining finishes at the cost of sliding door fixing agency.

The measurement for steel sliding door shall be correct to cm and area calculated in sq meters correct to two decimal places. The height of the shelter shall be measured from outside to outside the guide rail and width outside to outside of shutter including vertical position channels in sides when closed.

10.21 ROLLING SHUTTERS

Rolling shutters are to conform to IS:6248. They shall be suitable for fixing in specified position as in the drawing and shall include handles, locking arrangements etc. They shall be of specified type viz. push and pull type; operated with mechanical devices supplied by the manufacturers, reduction gear operated by mechanical device with chain or handle etc.

The shutters shall conform to approved specification. The spring shall be coiled type, manufactured from high tensile spring steel wire or strip of strength conforming to IS:4454 part 1-1981. The roller material shall conform to IS:1161.

The material, dimension, specification etc. of suspension shaft, supporting brackets, clamps, pulleys, pipe shaft, springs, ball bearings. Casings, guide channels cover, lock plates, sliding bolts, anchoring rods etc. shall conform to relevant IS and shall be conforming to specification in the nomenclature of the work.

The arrangement for fixing of rolling shutter in different situations in the opening shall be as per IS:6248-1979. The fixing shall be done accurately in a workman like manner that the operation of the shutter is easy and smooth.

The measurement for the rolling shutter shall be clear width and clear height of opening correct to a mm. The clear distance between the two jambs of the opening shall be clear width and clear distance between the sill and the soffit (bottom of lintel) of the opening shall be the clear height.

10.22 ROLLING GRILLS

Rolling grills are similar in design, construction, and operation of rolling shutter and all provisions of rolling shutter above shall be applicable to rolling grills except in respect of the shutter portion which shall conform to IS:6248.

A rolling shutter may have a rolling grill portion at top or bottom or at both places. The total height of all segments of grill portions shall not exceed 1 m and individual segment height shall not be more than 0.5m. The measurement of rolling grills shall be similar to rolling shutter.

10.23 MEASURE FOR STEEL WORK

(a) Steel work in single sections/built up section

Steel as fixed in position shall be measured in running metre correct to millimeter and weight calculated on the basis of standard tables correct to nearest kilogram. Unless otherwise specified weight of cleats, brackets, packing pieces, bolts, nuts washers, distance pieces, separators, diaphragm gussets (taking over all square dimensions) fish plates etc. shall be added to the weight of respective items. No deductions shall be made for rivet or bolt holes if its area is not exceeding 0.02 sqm. The standard weight of steel sections as given in relevant Indian Standard shall be used.

10.24 RATE

The rate shall include cost of all materials and labour involved in the job.

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